THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The slope of the line through $(4, 7)$ and $(5, -2)$ is: 1

(a) -8 **(b)** -10 **(c)** -7 **(d)** -9

2. Take $\pi = \frac{22}{7}$. The curved surface area of a cone of radius 14 cm and slant height 50 cm is: 1

(a) 2642 cm^2 **(b)** 2200 cm^2 **(c)** 1979 cm^2 **(d)** 2421 cm^2

3. A point on the x-axis at distance 6 units to the right of the origin has coordinates: 1

(a) $(0, 6)$ **(b)** $(7, 0)$ **(c)** $(6, 0)$ **(d)** $(-6, 0)$

4. The line $y = 3x + 1$ passes through $(2, p)$. The value of p is: 1

(a) 9 **(b)** 7 **(c)** 8 **(d)** 6

5. The sum of a rational number and an irrational number is: 1

(a) rational **(b)** an integer **(c)** irrational **(d)** zero

6. By identity, $103^2 =$ 1

(a) 10608 **(b)** 10610 **(c)** 10611 **(d)** 10609

7. The chord AB subtends 66° at the centre. The angle it subtends at a point on the minor arc is: 1

(a) 167° **(b)** 157° **(c)** 147° **(d)** 137°

8. A wheel of radius 28 cm (take $\pi = \frac{22}{7}$) makes one full turn. The distance covered (in cm) is: 1

(a) 175 **(b)** 178 **(c)** 177 **(d)** 176

9. Three coins are tossed together. The probability of getting exactly one head is: 1

(a) $\frac{3}{8}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{5}{8}$ **(d)** $\frac{1}{2}$

10. The 15th term of the AP $5, 8, 11, 14, \ldots$ is: 1

(a) 49 **(b)** 47 **(c)** 48 **(d)** 46

11. The number of factors of 36 is: 1

(a) 11 **(b)** 10 **(c)** 8 **(d)** 9

12. 60 mL of a 15% sugar solution is mixed with 120 mL of a 33% sugar solution. The concentration of the mixture is: 1

(a) 29% **(b)** 27% **(c)** 28% **(d)** 26%

13. The value of $146 \bmod 3$ (remainder on dividing 146 by 3) is: 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 3

14. The diagonals of a parallelogram intersect at O. If $OA = (2x + 8)$ cm and $x = 3$, then the diagonal AC is: 1

(a) 30 cm **(b)** 28 cm **(c)** 32 cm **(d)** 42 cm

15. The slope of the line $x - y = 6$ is: 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 0

16. Take $\pi = \frac{22}{7}$. The volume of a cone of radius 10.5 cm and height 14 cm is: 1

(a) 1455 cm^3 **(b)** 1779 cm^3 **(c)** 1617 cm^3 **(d)** 1942 cm^3

17. If $x < 0$ and $y > 0$, the point (x, y) lies in: 1

(a) Quadrant II **(b)** Quadrant I **(c)** Quadrant III **(d)** Quadrant IV

18. The graph of $y = 2x - 8$ cuts the x-axis at: 1

(a) $(-4, 0)$ **(b)** $(0, 4)$ **(c)** $(5, 0)$ **(d)** $(4, 0)$

19. A rational number $\frac{p}{q}$ equal to $0.1\overline{6}$ is: 1

(a) $\frac{1}{3}$ **(b)** 0 **(c)** $\frac{1}{7}$ **(d)** $\frac{1}{6}$

20. $x^2 + 8x + 16$ is the square of: 1

(a) $(x - 4)$ **(b)** $(x + 4)$ **(c)** $(x + 8)$ **(d)** $(x + 16)$

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Find the points where the line $4x + 6y = 36$ cuts the x-axis and the y-axis. 2

22. Take $\pi = \frac{22}{7}$. A cone has radius 7 cm and height 24 cm. Find its slant height, curved surface area and volume. 2

23. Find the midpoint of the segment with endpoints $(-3, 7)$ and $(9, 1)$. 2

24. Find $p(2)$ and $p(-2)$ for $p(x) = 4x + 3$. 2

25. Write $\frac{1}{6}$ as a decimal. 2

26. Find 104^2 using an identity. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Express $0.1\overline{6}$ in the form $\frac{p}{q}$. 3

28. Factorise $x^2 + 8x + 12$. 3

29. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 10)^\circ$ and $\angle C = (3x - 5)^\circ$. Find x and $\angle A$. 3

30. Find the area of the quadrilateral with sides 4, 5, 7, 10 cm that is inscribed in a circle. 3

31. In a survey of 200 students, 80 take the bus, 50 cycle and the rest walk. Find the probability that a randomly chosen student 3

(i) walks,

(ii) does not take the bus.

OR

Find the sum of the first 15 terms of the AP $6, 11, 16, 21, \ldots$. 3

32. Show that the statement "Every prime number is odd" is false by giving a counterexample. 3

OR

The mean of 10 observations is 53. A new observation 64 is added. Find the new mean. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. $ABCD$ is a cyclic quadrilateral with $\angle A = 75^\circ$, $\angle B = 88^\circ$ and side BC is produced to E, side AB to F. Find $\angle C$, $\angle D$, $\angle DCE$ and $\angle DAF$. 5

34. A rectangular lawn 30 m $\times$ 20 m has a path 2 m wide all around it, outside the lawn. Find the area of the path. 5

35. A letter is chosen at random from the word MISSISSIPPI. Find the probability that it is 5
(i) a vowel,
(ii) the letter S,
(iii) a letter from the first half of the alphabet (A to M).

36. Rohan saves ₹300 in the first month and ₹50 more each month than in the previous month. 5
(i) How much does he save in month 12?
(ii) How much does he save in 12 months in all?

OR

Prove that the square of an odd integer is odd and write the converse. Is the converse true? Test it with 7. 5

37. A family spent ₹2700 in January and ₹4000 in February. Food took 57% of the January total and 49% of the February total (100% stacked bars). In which month was more money spent on food, and by how much? 5

OR

Trace: s = 0; while n > 0, add (n mod 10) to s and replace n by its quotient on division by 10. Start with n = 5696. Write the digits picked and the final s. 5

38. The diagonals of a quadrilateral ABCD are AC = 17 cm and BD = 15 cm. P, Q, R, S are the midpoints of AB, BC, CD, DA. Show that PQRS is a parallelogram and find its perimeter. 5

OR

The relation between Fahrenheit F and Celsius C is $F = \frac{9}{5}C + 32$. Find F when $C = 50$, find C when $F = 212$, and say what the slope $\frac{9}{5}$ means. 5

END OF QUESTION PAPER — Total: 80 Marks

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